

VeriLib Atoms Proposal

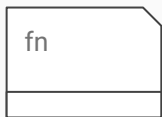
Atom Design

Different ways to present different information

Type of information	Way of presentation
Kind	Label at top left
Classification (optional)	Border color
Verification status	Color bar at the bottom
Compilation status	Color dot at top right corner

Kind

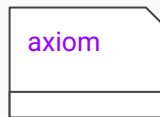
Example of how we can use (colored) labels like “fn”, “def”, etc to differentiate these 4 main “kind”:



Rust “exec”



- Verus “spec”
- Lean “def”, “abbrev”, “class”, “structure”, “inductive”, “instance”, “opaque”, “quot”



Lean “axiom”

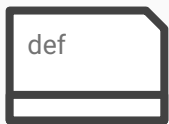


- Verus “proof”
- Lean “theorem”

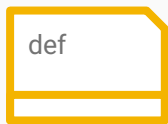
Note: We can also use shapes/icons/etc to differentiate “kind”.

Classification (optional)

Example of how we can use border colors to differentiate classes:



Scheme



Construction



Correctness



Security

Verification status

(for Rust ``exec`` atoms only)



Color	Condition
White	Default fallback color (all atoms will be colored white first before being colored appropriately if needed)
Grey	Verus unspecified (is-disabled: true), or Aeneas not translated
Yellow (<i>Aeneas only</i>)	translated or Aeneas-generated, but unspecified
Orange	specified, proof incomplete
Light Green	specified, "verified"
Dark Green	specified, "transitively-verified"

Compilation status

(for all other atoms)

Example of how we can use green/red dot at the top right corner:



Compilation success



Compilation with warning



Compilation failure

Arrow Design

Dependency arrows between atoms

TBC

Any questions/disagreements?